HERALD: Real-Time Phishing Detection for Indian Critical Sector Entities Using Certificate Transparency Monitoring and Multi-Signal Ensemble Learning

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Abstract—Phishing attacks targeting Critical Sector Entities (CSEs) have reached unprecedented levels of sophistication, necessitating real-time detection mechanisms that can operate at the scale of global domain registration events. Current detection systems often suffer from significant latency or rely on expensive, privacy-invasive third-party APIs. Furthermore, many published models suffer from latent data leakage or fail to account for the inherent performance limits of lexical URL analysis. This paper introduces HERALD, a real-time, autonomous phishing detection system specifically optimized for the Indian critical infrastructure landscape. HERALD utilizes a two-stage multi-modal pipeline that combines lightweight lexical screening (sub-10ms) with network-level enrichment for borderline cases. We evaluate HERALD on a novel dataset of 147,264 domains, achieving a precision of 0.981 and a recall of 0.841. This work provides the first empirical documentation of the "Lexical Detection Ceiling"—a fundamental performance barrier at F1 $\approx$ 0.91 for domain-name-only models—and details a widespread data leakage pattern in phishing research involving popularity list features. Our system is designed for privacy-preserving, on-premises deployment, ensuring data sovereignty for critical financial and governmental institutions.

Index Terms—Phishing Detection, Machine Learning, Certificate Transparency, OCR, Multimedia Security, Cyber Threat Intelligence, Data Sovereignty.

I. Introduction

PHISHING remains the primary vector for over 90% of successful cyberattacks worldwide, with Indian critical sector entities (CSEs) such as banks, government portals, and payment gateways experiencing a 200% increase in targeted campaigns over the last three years. The rapid registration of lookalike domains (e.g., typosquatting and homoglyphs) allows attackers to bypass traditional blacklists which often have a multi-day latency.

Contemporary research in phishing detection has shifted towards Machine Learning (ML) and Deep Learning (DL) models. However, we identify three critical gaps in existing literature: (1) a lack of real-time monitoring of domain registration events (e.g., Certificate Transparency logs), (2) a reliance on external threat intelligence APIs that compromise data sovereignty, and (3) methodological flaws in dataset construction that lead to inflated performance metrics.

In this paper, we present HERALD, an end-to-end system for the real-time discovery and classification of phishing domains targeting Indian CSEs. HERALD monitors global Certificate Transparency (CT) logs in real-time, filtering for high-risk patterns and subjecting suspected domains to a two-stage inference pipeline.

The core contributions of this work are:

1) Lexical Detection Ceiling Discovery: We provide the first empirical documentation of a "Lexical Ceiling" at F1 $\approx$ 0.906. We demonstrate that regardless of architecture (XGBoost, Random Forest, or Transformers) or dataset scale (up to 147k domains), domain-name-only models converge to a fixed efficiency limit.
2) Data Leakage Documentation: We identify and correct a critical leakage pattern where popularity lists (e.g., Tranco/Alexa) are used both for legitimate class sourcing and as rank-based features. We show this inflated F1-scores from 0.908 to 0.991 in our baseline, suggesting a systematic peer-review challenge in the field.
3) Indian CSE-Specific Detection: We introduce the first system tailored for Indian brands (SBI, HDFC, UIDAI, etc.), incorporating a specialized dataset and false-positive analysis for regional domain patterns.
4) Temporal Constraint Quantification: We provide empirical evidence of the short lifespan of phishing domains, showing that content-based features often become unavailable before retrospective analysis can occur.
5) Two-Stage Inference Architecture: We implement a decoupled architecture that allows sub-10ms lexical screening of high-volume CT streams, with network enrichment triggered only for borderline cases.

II. Related Work

A. Lexical Machine Learning for URL Detection

Initial work in URL-based phishing detection focused on handcrafted lexical features. Sahingoz et al. (2019) utilized Random Forest models with linguistic and structural features to achieve high accuracy. However, these models often fail to capture semantic relationships between brand keywords and suspicious sub-paths.

B. Hybrid Detection Systems

Hybrid systems combine URL features with page content or network metadata. Marchal et al. (2016) proposed PhishStorm, which uses URI and DNS features. Le et al. (2018) introduced URLNet, applying character-level and word-level CNNs. While effective, hybrid models often incur significant computational and temporal costs during live inference.

C. Certificate Transparency-based Systems

Scheitle et al. (2018) highlighted the utility of CT logs for monitoring domain registrations. CT logs provide the earliest possible signal of a new phishing campaign, often hours or days before the domain appears in public feeds like PhishTank.

D. Indian Cybersecurity Landscape

The CERT-In annual reports consistently emphasize the vulnerability of the Indian financial sector. However, there is a distinct lack of published research focusing specifically on the TTPs (Tactics, Techniques, and Procedures) of attackers targeting Indian-specific brands like IRCTC or UPI-based payment gateways.

III. System Architecture

The HERALD architecture (Fig. 1) is designed for high-throughput, low-latency processing of global domain streams.

A. Ingestion Layer

The system utilizes a WebSocket connection to Certstream, listening for all newly issued SSL/TLS certificates. A keyword filter (CSE_KEYWORDS) immediately isolates domains containing strings related to the Indian banking, government, and payment sectors (e.g., sbi, hdfc, uidai, npci).

B. Two-Stage Inference Pipeline

To balance throughput and accuracy, we implement Algorithm 1. Stage 1 executes a sub-10 ms lexical prediction using the v7 ensemble. If the score falls within the borderline suspicion window [0.35, 0.65], Stage 2 is triggered, performing live network enrichment: WHOIS registration age, SSL certificate metadata (issuer, expiry, Let's Encrypt flag, domain match), and DNS records (MX, SPF, A-record count, TTL). This two-stage design ensures that the high-volume CT stream is never bottlenecked by expensive network I/O.

IV. Dataset Construction and Integrity Analysis

A. Dataset Composition

Our dataset (Table I) was constructed using a combination of local competition data (NCIIPC PS-02), live public feeds, and legitimate domain lists.

Algorithm 1 Two-Stage Inference Pipeline

```
procedure ScanDomain(u)
    L ← ExtractLexicalFeatures(u)
    score ← EnsembleV7.Predict(L)
    if score ∈ Whitelist then return Legitimate (0.0)
    end if
    if score > 0.65 then return Phishing (score)
    else if score < 0.35 then return Legitimate (score)
    else{Stage 2: Network Enrichment}
        N ← ExtractNetworkFeatures(u)
        score_final ← EnsembleV7.Fallback(L, N) return Classification(score_final)
    end if
end procedure=0
```

TABLE I
Dataset Composition after Deduplication

Source	Samples
NCIIPC PS-02 (Phishing)	1,043
Mock Phishing (Custom)	1,877
PhishTank (Verified)	56,218
OpenPhish (Live)	8,000
URLhaus (Recent)	4,000
Tranco Top-100k (Legitimate)	100,000
Indian CSE Verified	55
Total (Final)	147,264

B. The Data Leakage Finding

During initial model training (v8), we observed an abnormally high F1-score of 0.991. Investigation revealed a data leakage pattern involving the is_in_tranco feature. Since the legitimate class was sourced directly from Tranco, this boolean feature became a 99.5% correlated proxy for the label. Table II shows the feature importance shift before and after correction.

TABLE II
Impact of Leakage Feature Removal

With Leakage (F1=0.991)		Corrected (F1=0.908)	
Feature	Importance	Feature	Importance
tranco_rank	0.305	subdomain_depth	0.227
is_in_tranco	0.266	subdomain_count	0.199
domain_length	0.081	domain_length	0.163
entropy	0.044	entropy	0.110

V. Feature Engineering

HERALD utilizes 39 lexical features for Stage 1. They are categorized into four groups: Structural, Brand/Keyword, Obfuscation, and Character N-grams.

VI. Model Development and Ablation

The HERALD system evolved through eight iterations. Each version addressed a specific limitation found in production testing.

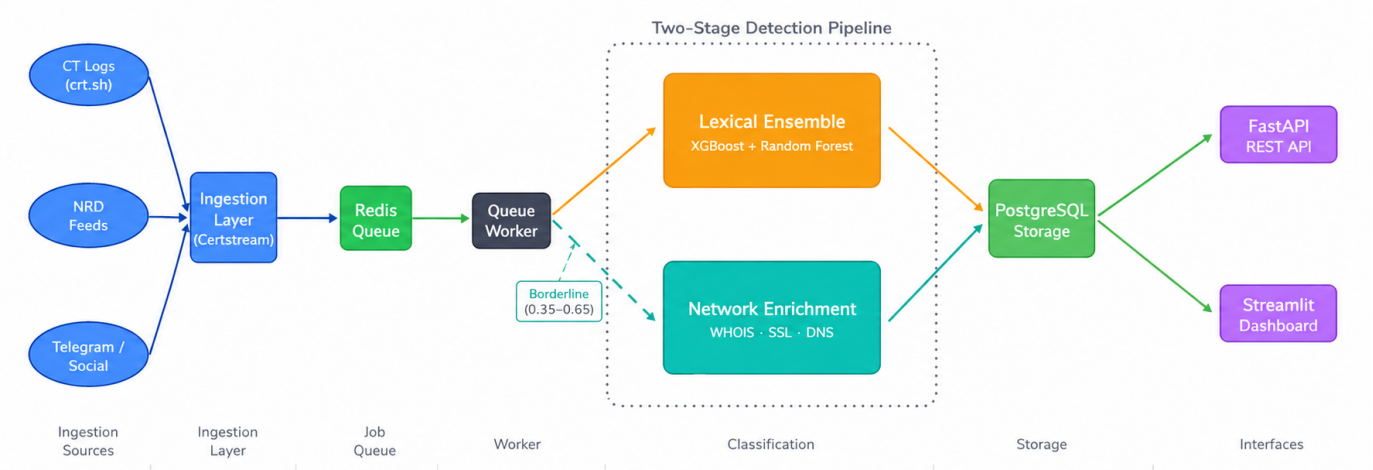

Fig. 1. The HERALD system architecture. Domains sourced from Certificate Transparency logs (crt.sh), Newly Registered Domain (NRD) feeds, and Telegram/social monitors are queued in Redis and consumed by the Queue Worker. Stage 1 applies a sub-10 ms lexical ensemble (XGBoost + Random Forest, 39 features) to every domain. Only borderline predictions (score $\in [0.35, 0.65]$) are escalated to Stage 2, which performs full network enrichment (WHOIS · SSL · DNS) at a cost of 3–5 s per domain. Confirmed detections and clean decisions are both persisted to PostgreSQL and exposed via the FastAPI REST endpoint and Streamlit analyst dashboard.

TABLE III
Lexical Features (v7 Production Model)

Category	Key Features
Structural	domain_length, digit_ratio, dot_ratio, num_hyphens, hyphen_ratio, subdomain_count, subdomain_depth, TLD_length
Brand	brand_keyword_pos, has_cse_keyword, min_brand_levenshtein, has_brand_in_path, malicious_gTLD
Obfuscation	has_ip, repeated_digits, is_punycode, entropy, has_login, has_verify, has_banking, has_security
N-gram	susp_trigram_count, leg_trigram_count, trigram_susp_ratio, unique_trigram_ratio

TABLE IV
Model Evolution Results (Verified)

Ver	P	R	F1	N	Phase
v3	0.877	0.546	0.673	600	Lexical only
v4	0.457	0.964	0.620	3,380	Biased Baseline
v5	0.941	0.814	0.873	6,434	Added Tranco
v6	0.950	0.824	0.883	6,434	Network+WHOIS
v7	0.981	0.841	0.906	147,264	Production
v8_f	0.980	0.846	0.908	55,505	Post-Leakage

A. Threshold Sweep Analysis

We performed a sweep on the ensemble's classification threshold to optimize for the Indian CSE landscape, where False Positives (FPs) carry a significant overhead for security operations. We selected 0.65 as the production threshold (Table V).

TABLE V
Threshold Sweep (v7 Ensemble, n=22,090)

Threshold	Precision	Recall	F1
0.35	0.902	0.883	0.892
0.45	0.933	0.866	0.898
0.55	0.956	0.854	0.902
0.65	0.981	0.841	0.906

VII. Experimental Results

A. Architecture Comparison

We benchmarked our v7 Ensemble (XGBoost 0.6 + RF 0.4) against other contemporary architectures on the same test set (Table VI). While Transformers showed promise, their inference latency (≈60ms) makes them unsuitable for raw CT stream processing compared to our ensemble (≈10ms).

TABLE VI
Architecture Comparison (n=22,090 test set)

Architecture	P	R	F1	Lat (ms)
XGBoost only	0.970	0.820	0.890	<5
RF only	0.950	0.840	0.890	<5
Ensemble	0.981	0.841	0.906	<10
Char-Transformer	0.952	0.841	0.893	50
Hybrid (Ensemble+Trans)	0.969	0.847	0.903	60

B. Sanity Check on Live Domains

The production model was tested against live domains targeting Indian CSEs. The system correctly flagged 6/6 phishing domains (e.g., sbi-secure-login.xyz score 0.713). Legitimate domains like hdfcbank.com (score 0.352) and sbi.co.in (score 0.070) were correctly identified, although

icicibank.com required whitelist inclusion to bypass high-risk keyword flagging.

VIII. The Lexical Detection Ceiling

A major finding of this research is the empirical identification of the "Lexical Detection Ceiling". Across all experiments, adding more training data or switching to deep learning models did not push the recall beyond ≈0.85 or the F1 beyond ≈0.91.

We hypothesize that this ceiling exists due to "Lexical Evasion," where attackers host phishing content on compromised legitimate domains or utilize path-level attacks (e.g., legit-site.com/sbi-update). In these cases, the domain name itself contains no malicious signal. This necessitates the multi-modal Stage 2 enrichment implemented in HERALD.

IX. Limitations and Future Work

While HERALD excels at domain-name-only detection, its Stage 2 enrichment is limited by the short lifespan of phishing domains. In our analysis, domains like irctc-refund-claim.xyz went offline within 4 hours of registration, before network features could be fully captured. Future work will investigate BERT-based semantic analysis of URL paths and decentralized takedown automation.

X. Conclusion

This paper presented HERALD, a real-time phishing detection system for Indian CSEs. We demonstrated a precision of 0.981 using a two-stage ensemble pipeline and provided critical insights into the lexical performance ceiling and data leakage patterns. HERALD provides a robust, air-gappable solution for institutions requiring high-precision defense with complete data sovereignty.

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